

Credit Risk Intelligence Platform

An AI-powered platform for loan default prediction, explainability, and natural-language data access



Building a bank-ready credit risk platform

Multi-table data

Joined application, bureau, prior-loan, and monthly repayment history — not just a single table

Explainable by design

Every prediction is backed by SHAP reasoning and auditable business rules, not a black box

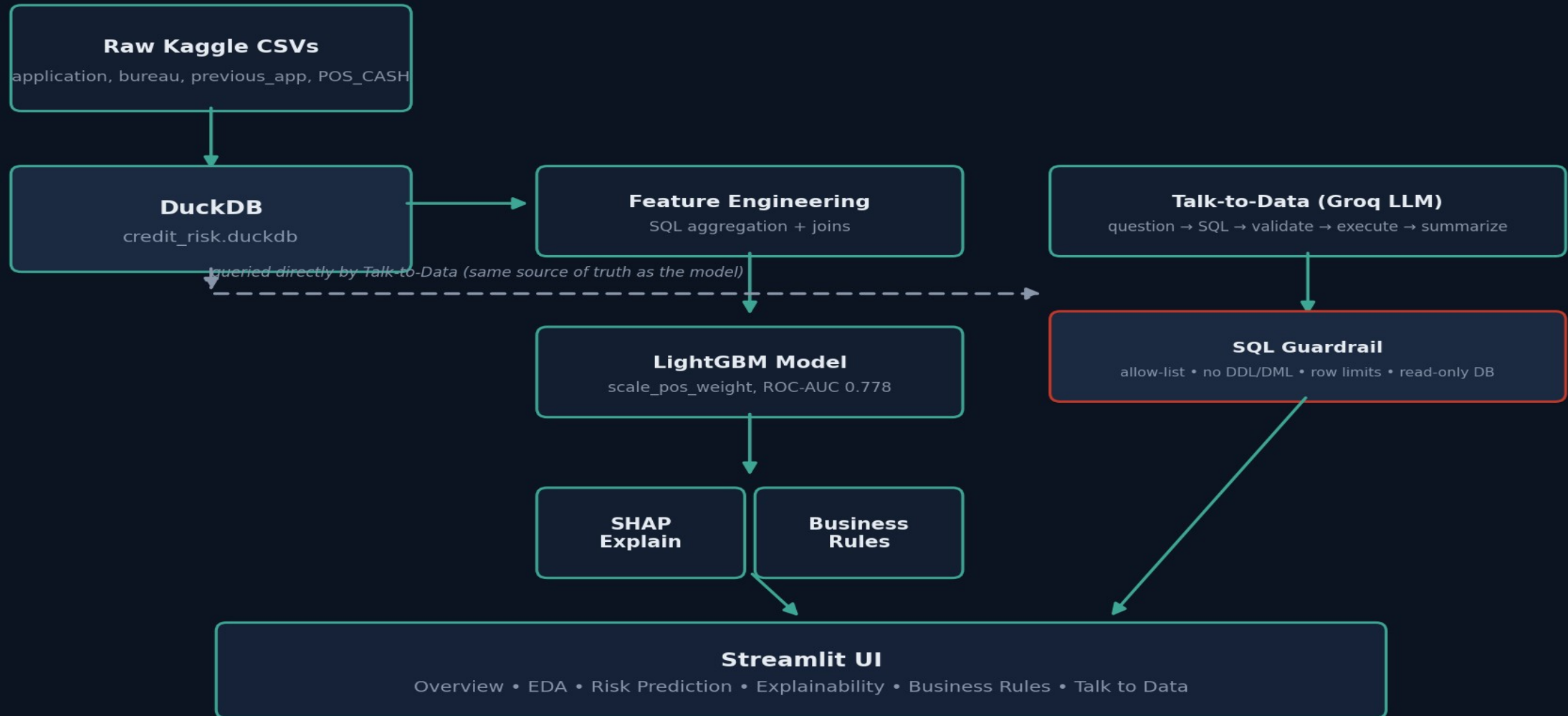
Talk to the data

A guardrailed NL-to-SQL chatbot for non-technical stakeholders to explore the data directly

Ship it

Fully Dockerized, single-command deployment

How the pieces fit together



Multi-table feature engineering, not just one CSV

Most submissions on this dataset use application_train.csv alone. This platform joins five additional tables — aggregated in SQL for speed on tables up to 27M rows — to capture signal a single-table model would miss.

applications

307,511 rows

Core applicant demographics & loan terms

bureau + bureau_balance

1.7M + 27.3M rows

External credit history & monthly DPD status

previous_applications

1.7M rows

This applicant's prior loans with the same lender

pos_cash_balance

10M rows

Monthly days-past-due (DPD) on cash/POS loans

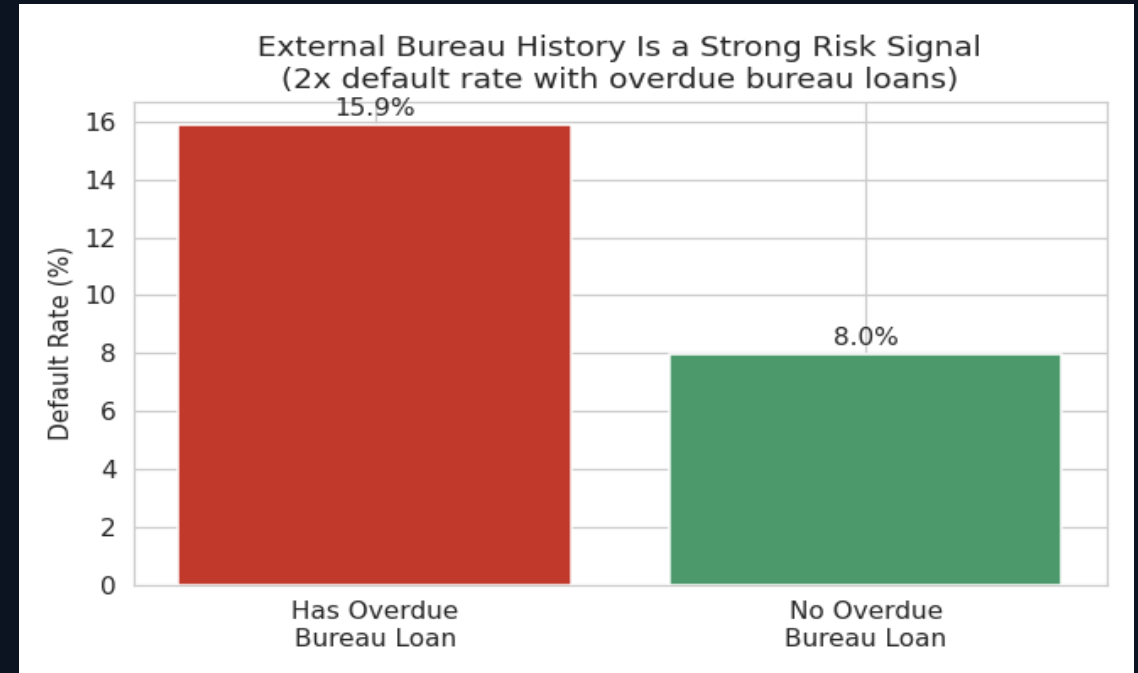
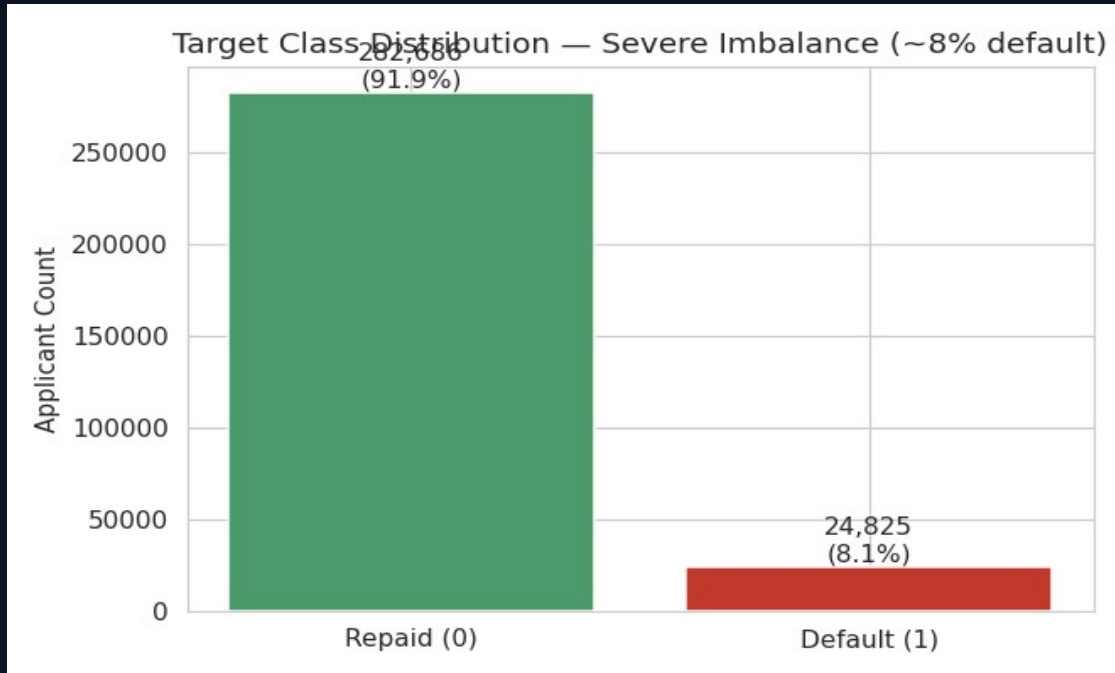
credit_card_balance

3.8M rows

Monthly card balance & utilization — new default-risk signal

Result: 165 features across 6 tables, feeding a single unified model.

What the data says before any model is built



- Only 8.07% of applicants default — a naive "approve everyone" model would be 92% "accurate" and useless
- An overdue external bureau loan nearly doubles default risk (15.9% vs 8.0%)

LightGBM with an evidence-based imbalance strategy

ROC-AUC

0.780

PR-AUC

0.277

Recall @ 0.5

66.7%

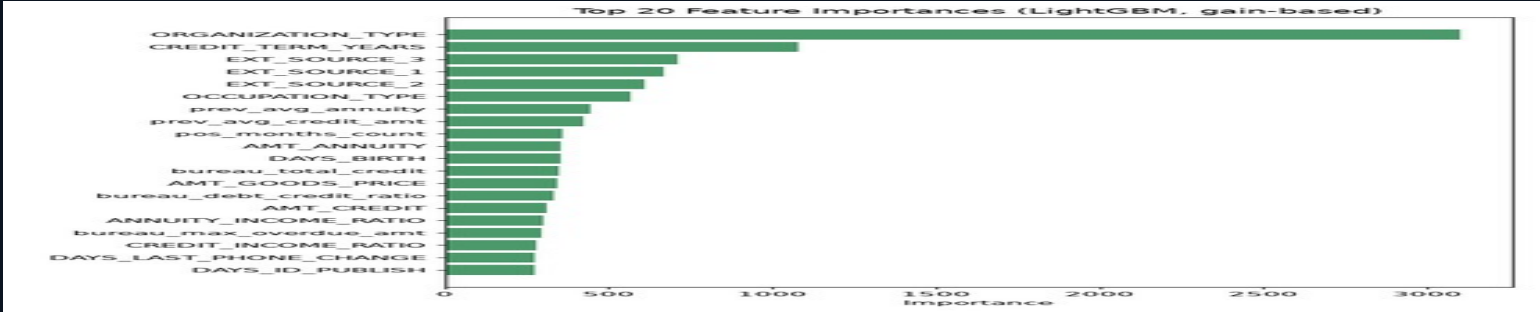
Features

165

Class imbalance (~8% default rate): scale_pos_weight vs SMOTE — tested, not assumed

Approach	Validation ROC-AUC	Notes
scale_pos_weight (production)	0.7804	Simpler, faster, no synthetic data
SMOTE (0.5 ratio)	0.7792	Statistically indistinguishable (difference = 0.001)

Evaluated on ROC-AUC / PR-AUC, not accuracy — accuracy is misleading under 8% class imbalance.



SHAP: auditable reasons behind every prediction

- TreeExplainer — exact SHAP values for gradient-boosted trees, not an approximation
- Per-applicant: top features, direction, and a plain-English summary
- Global: beeswarm plot across a 2,000-applicant sample
- Example: correctly flagged 22 active bureau loans + 70.7% refusal rate on a genuinely-defaulted applicant (p=0.955)

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◆ Credit Risk Platform

AI-powered credit risk intelligence — NeoStats assignment

Overview

Data Exploration (EDA)

Risk Prediction

Explainability

Business Rules

Talk to Data

Deploy

Explainable AI

SHAP-based explanation of individual predictions — auditable, feature-level reasoning.

High Risk

Default probability: 35.8%

Summary: Risk was pushed UP mainly by: amt goods price, organization type, credit term years. Risk was pulled DOWN mainly by: own car age, ext source 1, bureau debt credit ratio.

Top contributing factors:

▼

OWN_CAR_AGE = 9.0 (decreased risk, impact 0.180)

▲

AMT_GOODS_PRICE = 450000.0 (increased risk, impact 0.178)

▼

EXT_SOURCE_1 = 0.5 (decreased risk, impact 0.128)

▲

ORGANIZATION_TYPE = Business Entity Type 3 (increased risk, impact 0.124)

▼

bureau_debt_credit_ratio = 0.22343412178294153 (decreased risk, impact 0.121)

▲

CREDIT_TERM_YEARS = 1.6666666666666667 (increased risk, impact 0.114)

▲

CODE_GENDER = M (increased risk, impact 0.113)

▼

bureau_max_overdue_amt = 0.0 (decreased risk, impact 0.090)

Global Feature Importance

How each feature affects predictions across the whole applicant population.

Bridging ML output and credit policy

- Bins top-25 model features and surfaces any bin where default rate is materially above baseline
- Every rule backed by measured lift + support (n), not just model-internal importance
- Auditable by a policy team independent of trusting the ML model
- Example: Low-skill Laborers default at 2.12x baseline (n=2,093)

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○ Overview

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○ Explainability

● Business Rules

○ Talk to Data

Deploy

:

Derived Business Rules

Plain-English underwriting rules mined from the model's learned patterns, each backed by measured lift on real validation data — not just model-internal importance.

If OCCUPATION_TYPE == 'Low-skill Laborers'

→ default rate 17.2% vs 8.1% baseline

2.12x lift

(n=2,093)

If ORGANIZATION_TYPE == 'Transport: type 3'

→ default rate 15.8% vs 8.1% baseline

1.95x lift

(n=1,187)

If EXT_SOURCE_3 in [-0.00, 0.37]

→ default rate 15.1% vs 8.1% baseline

1.87x lift

(n=61,993)

If EXT_SOURCE_2 in [-0.00, 0.39]

→ default rate 14.3% vs 8.1% baseline

1.77x lift

(n=76,716)

If EXT_SOURCE_1 in [0.01, 0.33]

→ default rate 13.6% vs 8.1% baseline

1.68x lift

(n=33,534)

If ORGANIZATION_TYPE == 'Construction'

→ default rate 11.7% vs 8.1% baseline

1.45x lift

(n=6,721)

If ORGANIZATION_TYPE == 'Restaurant'

→ default rate 11.7% vs 8.1% baseline

1.45x lift

(n=1,811)

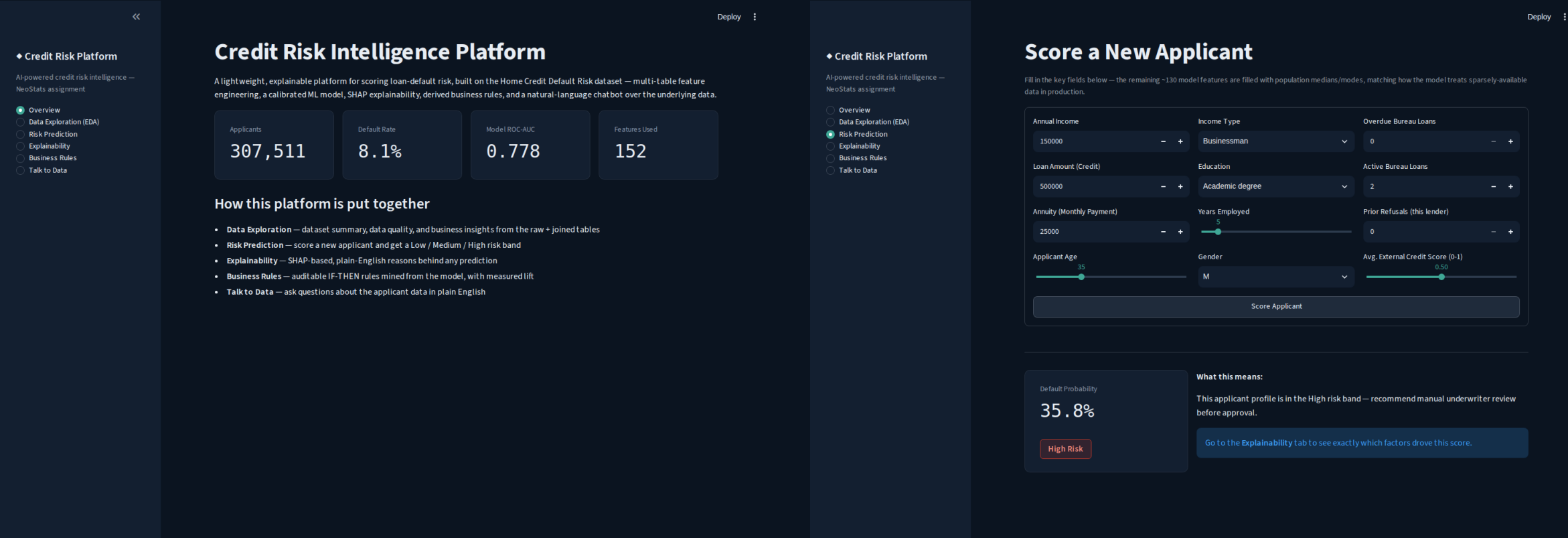
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NL-to-SQL with guardrails enforced in code

- Two-stage LLM calls: SQL generation (temp 0.1) → grounded answer summarization from real returned rows
- SQL validator blocks: DDL/DML keywords, chained statements, hallucinated tables/columns
- Auto row-limiting + read-only DB connection as a second defense layer
- Tested against real adversarial inputs (DROP TABLE, injection, fake tables) — all correctly blocked
- 7 verified working query patterns (exceeds the required 5)

The screenshot displays a web application interface for a 'Credit Risk Platform'. On the left, a sidebar contains a navigation menu with options: Overview, Data Exploration (EDA), Risk Prediction, Explainability, Business Rules, and Talk to Data (which is selected). The main content area is titled 'Talk to Your Data' and includes a subtitle: 'Ask questions in plain English — answered by an LLM-generated, validated SQL query against the real applicant database.' Below this, there is a prompt 'Try one of these, or type your own question:' followed by six example questions in buttons: 'What is the average income of applicants w...', 'How many applicants have more than 2 over...', 'What's the default rate for applicants with a...', 'Show me the top 10 organization types by a...', 'How many applicants were previously refus...', and 'What is the average credit amount for fema...'. A text input field labeled 'Your question' is provided for user input, with an 'Ask' button below it. In the top right corner, there is a 'Deploy' button and a settings icon.

One platform, five sections, real data throughout



Overview (KPIs from real data)

Risk Prediction (live scoring + risk band)

Streamlit • dark navy/teal design system • IBM Plex Sans • responsive KPI cards

One command to run the full platform

1

Clone & configure

git clone the repo, copy .env.example → .env, add a free Groq API key

2

Prep data & model

Download 5 Kaggle CSVs → run loader, preprocessor, train, evaluate, rules scripts

3

docker-compose up

Builds the image and launches Streamlit on port 8501 — single command

Dockerfile • docker-compose.yml • .env.example — all in the repo root, per the required structure

github.com/malavika-krishna24/credit-risk-platform

Known limitations & what's next

installments_payments.csv not joined

Repayment-timing signal is already substantially covered by POS_CASH_balance and credit_card_balance's own DPD tracking

Risk-band thresholds are defaults

Low/Medium/High cutoffs should be tuned against a real bank's approval/loss economics in production

Prediction form uses ~12 key fields

Remaining ~153 features filled from population medians/modes — a deliberate UX tradeoff for a demo interface

No monitoring/retraining pipeline

A production deployment needs drift detection and scheduled retraining — out of scope here

Thank you

Credit Risk Intelligence Platform — built end-to-end with real data, tested at every stage

github.com/malavika-krishna24/credit-risk-platform